

A one-day seminar with Stakeholders on 17 April 2025

In Defence of Seed Sovereignty: Linking Community Seed Banks, Custodian Farmers, and Other Conservationists with the National Gene Bank

Seminar Recommendations

A policy structure on the entire structure and functioning of the second gene bank (SNGB) must be developed and implemented. The policy document should specifically address the following.

1. Define the conservation support to the private sector institutions and the roles of the public sector and private sector in the SNGB
2. Establishing a 'Governing Body (GB)' structured with representation from ICAR, Forest Department, BSI, SAUs, ICAR-NBPGR, NBA, SBBs, PPVFRA, Civil Society, and Community Seed Banks to ensure that the Global and Domestic legal Instruments are not violated, ABS enforced to establish SNGB as an effective autonomous functional unit. Private sector organizations and CSBs being the Members of the GB, arbitrary decisions going against their interests will not arise
3. National Indexing of the Germplasm in the linked CSBs and SNGB to be developed and made available in SNGB Domain as well as at every CSB.
4. Gaps of germplasm collection concerning crop plants, medicinal plants, tubers and forest plants need to be assessed systematically on a regular basis by the ICAR-NBPGR, discussed in SNGB GB to be followed up with the collection missions.
5. Germplasm Evaluation and Characterization must be undertaken on-farm under the jurisdiction of every CSB linked to the SNGB under the supervision of ICAR-NBPGR Scientists.
6. Elite germplasm lines and the farmers' varieties with specific traits of significance need to be identified and displayed in the SNGB portal as well as the corresponding CSBs portals to which the specific collections have the qualified jurisdiction. This will facilitate quick access by the users especially the industry facilitating CSB functional Units more dynamic and generate resources meeting the operational expenses of the CSBs to some extent.
7. *In-situ* and on-farm conservation of germplasm should be treated as a significant part of National Germplasm Conservation on par with *Ex-situ* conservation.
8. Financial and technical support should become part of budgetary allocation and regular Capacity building imparted to the working staff of the CSBs.
9. Community Seed Banks (CSBs) must be independent state-supported autonomous bodies for making decisions in multiplying and monetization' of local/farmers' and/or traditional varieties.
10. CSBs may be trained in marketing allowing them to join the supply chain as farmers' cooperative bodies
11. The data on germplasm accessed by the industry must be shared with the National Biodiversity Authority (NBA)?
12. Conservation support must also go to several civil societies working for the conservation. There are several civil societies like Gene Campaign and Sanjeevini in tribal areas with functional CSBs
13. A fool proof mechanism needs to be worked out to ensure 'Fair and Equitable Benefit sharing' Also What exactly is the Fair and Equitable share needs to be defined
14. The penalties for violation of the provisions of the Act and punishments must be enforced. The penalty must be a deterrent and discourage the violators from such indulgence.

Recommendations for unified seed policy for the country but with a bearing on the SNGB

15. Policy document to define the germplasm access to the private sector in the backdrop of new amendments and the ABS in tune with the CBD, BDA, and the Nagoya Protocol.
16. The policy document must define IPRs in the context of the private sector, the farmers, and farming communities as breeders and conservers.
17. PPVFRA registrations need to be monitored by ICAR, SAUs, CSIR as provided in the policy
18. Policy paralysis needs to be corrected concerning '**the farmers varieties**' accounting to nearly 20% of PPVFRA registrations belonging to 'Farmers'/Farmer Producer Organizations'. An 'FPO' in the context of PPVFRA Registrations need to be defined more explicitly.
19. A mechanism needs to be developed for DUS testing of Farmers' varieties facilitating registration and further assistance to multiply and market **farmers' varieties**. Farmer cooperatives and local seed companies may be encouraged to help the 'farmers' varieties for multiplication, marketing, etc. A Goldmine of about 2038 varieties registered by the farmers or FPOs is crying for the attention of stakeholders.
20. Revitalization of rainfed agriculture and seed diversity in the context of strengthening the community seed systems needs to be incorporated in the policy
21. The NBPGR Database must be made accessible to the general for information, tangible suggestions, stakeholder benefit and knowledge. The ICAR-NBPGR (National Bureau of Plant Genetic Resources) database can be used as prior art in patent applications.
22. Attempts to subvert the '**farmers' rights**' granted in the PPVFR Act should be blocked as the intent of those standing against the interests of the farming community ultimately wanting to take farming in India towards UPOV era.
23. Therefore, in light of the new gene bank initiative, farmers' varieties need to be given conservation support. They need to be supported by the government, compulsory licenses may be issued (CLs) to help the '**exceptionally good farmers' varieties**', lacking institutional support.

The Context

Union Budget for 2025-26, Finance Minister (FM) Nirmala Sitharaman announced the second 'National Gene Bank,' which will hold a million germplasm lines to be set up for future food and nutritional security. She mentioned that it is designed to support the public and private sectors in conserving genetic diversity.

During a post-budget webinar (video conferencing), Prime Minister Shri Narendra Modi announced that a Gene Bank will be established to conserve the country's genetic resources. This initiative aims to ensure genetic resources and food security for future generations (<https://pib.gov.in/PressReleasePage.aspx?PRID=2108488>).

A reminder of India's commitment to the global agreement and domestic legislation

India, a signatory to the Convention on Biological Diversity (CBD 1994), enacted the Biological Diversity Act in 2002, aiming to protect biodiversity, promote sustainable use, and ensure fair and equitable benefit sharing, fulfilling its international obligations and national priorities.

National Biodiversity Strategy and Action Plan (NBSAP): India has an NBSAP to guide its actions towards biodiversity conservation.

Commitment to Biodiversity Conservation: India recognizes the importance of biodiversity conservation, as an international obligation and a national priority, linking it to local livelihoods and sustainable development. According to the CBD, India enacted the Biological Diversity Act in 2002, and notified Biological Diversity Rules in 2004, to give effect to the provisions of this Convention, including those relating to ABS. The Act is implemented through a three-tiered institutional mechanism at the national, state, and local levels.

Compliance with Agreements and Acts

In India, there is a clear lack of enthusiasm and compliance with the Biological Diversity Act (2002) and the Nagoya Protocol (2010), which aim at three fundamental principles viz., Conservation, Sustainable Utilization, and Fair & Equitable sharing of benefits from genetic resources. There appears to be gross negligence in obtaining prior informed consent from local communities, neglecting benefit-sharing arrangements, and neglecting to seek prior approval from the National Biodiversity Authority for intellectual property rights based on Indian biological resources.

Goals of the Seminar: To focus on the following items and make recommendations

1. Structuring the model of a functional national gene bank and defining the vital role of community seed banks (CSBs) and the private sector
2. Emphasis on mission mode collection and evaluation of genetic diversity by public sector institutions led by ICAR-NBPGR
3. Capacity building to CSBs and the custodian farmers maintaining the genetic diversity towards promoting *in situ* conservation
4. To reiterate that access to germplasm is available to all stakeholders, including the private sector, in conformity with national rules and regulations
5. Evolving effective strategies of the ABS provided in BDA as well as the Nagoya Protocol
6. A governing body / A platform to be shaped in tune with our primary objective, "In Defense of Seed Sovereignty with a coherent team for coordination and leadership"

Participants: Scientists, communities involved in seed banks, civil societies, legal experts, subject matter specialists, researchers, students, seed industry, and general public.

Venue: Agri Biotech Foundation, PJTAU Campus, Rajendranagar, Hyderabad 500030

The following are the primary organizers and sponsors of the seminar

- | | |
|---|---|
| 1. Plant Protection Association of India (PPAI) | 3. All India Peoples' Science Network (AIPSN) |
| 2. Kisan Mazdoor Commission (KMC) | 4. Gene Campaign (GC) |

The following are the logistics supporters

- | | |
|---|-----------------------|
| 1. Centre for Sustainable Agriculture (CSA) | 3. Jana Vignana Vedik |
| 2. Deccan Development Society | |

Proceedings of the Seminar

Introduction and Context

The seminar titled "In Defence of Seed Sovereignty: Linking Community Seed Banks, Custodian Farmers, and Other Conservationists with the National Gene Bank" was held on 17 April 2025 in Hyderabad, a city that has long served as a vibrant hub for agricultural research and innovation. Hosting a cluster of prestigious institutions - including the State Agricultural University, ICAR institutes, national-level research organizations, and numerous seed companies - Hyderabad offered the perfect setting for deliberations on a theme as critical as seed sovereignty.

The event commenced with opening remarks by Dr. Soma Marla, who set the tone by highlighting the importance of seed sovereignty in safeguarding India's agricultural biodiversity and farmer autonomy. He emphasized the need to strengthen institutional linkages between community seed banks, custodian farmers, and national gene banks while ensuring fair and equitable benefit sharing among stakeholders.

Dr Soma Marla: The meeting took off with a welcome address by Dr Soma Marla, Former Principal Scientist of NBPGR. He spoke in detail about the context of the seminar, the significance of the seminar, and the diversity of experts invited and taking part in the deliberations. He spoke about the initiative of the Union Government on the Second National Gene Bank (SNGB), the existing First National Gene Bank (FNGB), and the reasons behind setting up the SNGB in terms of government view as well as the positive views of the civil society, the private sector and the public sector institutions.



In his welcome address, Dr. Marla underscored the urgency of establishing a second National Gene Bank, noting the commendable efforts of the National Bureau of Plant Genetic Resources (NBPGR) in this direction. He emphasized that the model and functioning of this new gene bank must be shaped by inclusive principles - particularly focusing on enriching the gene pool and ensuring entitlements and benefit-sharing for key stakeholders.

He outlined three broad categories of stakeholders whose contributions and rights must be recognized in the gene conservation ecosystem:

1. Public sector institutions,
2. Community seed banks and custodians, and
3. Private seed companies.

Highlighting the inclusive and collaborative spirit of the event, Dr. Marla welcomed a galaxy of stakeholders from across the national policy arena, who came together to engage in critical dialogue on fair benefit-sharing mechanisms across these sectors.

Dr Marla recalled the initiative and the efforts put in by Dr B Sarath Babu, President, Plant Protection Association of India and former Principal Scientist and Head of ICAR-NBPGR Regional Station< Hyderabad. Dr Babu worked incessantly to bring galaxy of scientists, policy makers, members of civil society, the industry and other stakeholders from various walks of life on to this platform. Reflecting on the sobering history of agrarian distress, especially the wave of farmer suicides that occurred in the early 2000s - most starkly in cotton-growing regions - Dr. Marla invoked the silent yet powerful presence of Dr. P. Sainath, whose journalism brought these crises to light. He also acknowledged the pioneering work of Dr. G. V. Ramanjaneyulu, a torchbearer of sustainable farming practices and a longstanding advocate for farmer rights. Dr Marla welcomed Shri Vadde Shobanadreesara Rao, Dr Kuldeep Singh, Dr Dinesh Abrol, Dr Kavitha Kuruganti, Dr Donti Narasimha Reddy, Dr Raghuram, Dr Gopakumar and Nitin Ramakrishna from TWN, Dr Venugopala Rao and all other towering personalities from various walks of public life.

Dr Suman Sahai: Dr Suman Sahai, Founder Chairperson of Gene Campaign. Gene Campaign works on national and international policy issues related to food, nutrition, agriculture, bioresources, agrobiodiversity, and climate change. It works with small farmers, especially women farmers to make their agriculture and food production more climate resilient and sustainable.



At the outset, Dr Suman stressed the need to recognize the contributions of farmers in the conservation of genetic diversity. She rightly pointed out that Conservation can't happen without the native farmers and farmer communities. The talk of conservation support to both the industry and public sector is worrisome because they left out the primary in situ on farm conservator communities. The state governments, union governments, scientific institutions, the least to the industry don't own the germplasm. It is owned by the people of this country more specifically the farmers and the farming communities. It is they who used it for centuries who conserved it, used it, developed it through breeding methods like selection and multiplication, and distributed among the communities adopting perfect scientific protocols. The saved the seed and the diversity of the germplasm through in situ on farm conservation. She therefore suggested that the discussions must be candid focus on policy interventions which will have influence on the proposed new gene bank and a retrospective effect on the existing gene bank.

She wants to know the track record of public sector in terms of access to genetic resources from the FNGB like how much is accessed by both the public and private sectors and how much is introduced into the bank. What products have been developed from the accessed material and how much benefit sharing has been made to the National Gene Fund and how much has been shared with the communities?

Now in the context of SNGB, the following questions are being flagged.

- a) What is the nature of this conservation support announced for the private sector?
- b) How the private sector is going to access the germplasm? Are they access from the farmers directly?
- c) Who is monitoring the access today?
- d) Does the conservation efforts of the farming communities in situ and on farm conservation not amounting to conservation?
- e) If so What kind of conservation support is being given to the communities and community seed banks for in situ conservation
- f) What kind of capacity building imparted to the community seed banks?
- g) Are the private sector institutions going to set up their own gene banks? Do they collect and put it in the SNGB?
- h) Access to germplasm is governed by the Access and Benefit Sharing! Is this protocol being honoured? Whether so far any ABS amounts received in to the Gene Fund?
- i) Whether the data on germplasm accessed by the industry shared with the National Biodiversity Authority (NBA)?
- j) Why in spite of the major responsibility of in situ conservation handled mostly and naturally by the communities and the community seed banks resulting in enrichment of gene banks (ex situ conservation), no financial support given to the community seed banks?

- k) Conservation support must also go to several civil societies working for the conservation. There are several civil societies like Gene Campaign and Sanjeevini working in tribal areas promoting community seed banks that are badly in need of support.
- l) What is the mechanism through which Fair and Equitable benefit sharing is decided? What is Fair and What is Equitable?
- m) The penalties for violation of the provisions of the Act and punishments must be enforced. The penalty has to be a deterrent and discourage the violators from such indulgence.

Dr Dinesh Kumar Abrol: Dr. Dinesh Abrol is a prominent scholar known for his extensive research and work in the field of Science and Technology Studies (STS) and Intellectual Property Rights. Dr. Abrol's work is characterized by a critical engagement with the social, economic, and political dimensions of science and technology, with a particular focus on issues relevant to developing countries.

Dr Abrol also very significantly highlighted the injustice meted out to the native conservators and communities conserving the germplasm and the Indigenous Traditional Knowledge. Several issues were flagged by Dr Abrol as follows in the seminar.

- a) Germplasm access to private sector in the light of new amendments and the benefit sharing in tune with the CBD, BDA and the Nagoya Protocol.
- b) He brought out the issue of IPRs in the context of private sector and the farmers and farming communities as breeders and conservers.
- c) He also highlighted the public sector today not being active in seed sector due to variety of reasons. In continuation of his position, he mentioned that until 2022, while private sector registered about 772 entries with the PPVFRA public sector as a whole registered only 165 entries. In terms of public sector share of registrations were hardly 3% while the rest was done by private, farmers and farmer producer organizations.
- d) While the private sector registered basically Essentially Derived Varieties, the public sector registered Extant Varieties.
- e) PPVFRA wanted the registrations to be monitored by ICAR, SAUs, CSIR but there seems to be no progress and hence no date generated in the context.
- f) The farmers' varieties accounting for 20% of registrations belonged to Farmer Producer Organizations which again essentially are related to institutions and companies in the name of FPOs promoted by the politicians.
- g) While expressing the technical hardships with respect to Distinctiveness, Uniformity & Stability (DUS) testing, he mentioned that there were about 2038 varieties registered in the name of farmers. Farmers also face challenges though they have some good varieties registered with PPVFRA, they can't multiply and market the varieties themselves. Seed industry will not have any interest basically because they are not hybrids and repeated sales year after year will not happen and therefore not profitable.
- h) Farmer cooperatives and local seed companies would have joined the efforts and helped the farmers varieties for multiplication, marketing and mutual benefit.
- i) Therefore in the light of new gene bank initiative, farmers' varieties need to be given conservation support. They need to be supported by the government, compulsory licenses may be issued (CLs) indicating the hardships for the exceptionally good farmers' varieties lacking institutional support.

- j) Community seed banks need to be established in the gaps or inactive areas of diversity and their enterprenurial capabilities enhanced.
- k) Dr Dinesh Abrol also spoke extensively about the compensation to the farming communities, involving farm women in large numbers in conservation activities etc.
- l) He passionately spoke about the experience of domestic legislations in the past two decades and the shortfalls concerning Access, Sustainable Utilization and ABS highlighted.
- m) The new gene bank initiative in his view is an opportunity to take stock of the situation there should be no time lapse to form the working groups as early as possible to reconstruct the policy frameworks.

Dr Kuldeep Singh: Dr. Kuldeep Singh is a prominent figure in plant science and agriculture, with a diverse background and expertise in various fields. He has held positions at major institutions like ICRISAT, PAU Ludhiana, and ICAR-NBGPR. His work spans plant breeding, molecular genetics, and genomics, with a focus on developing sustainable agricultural practices.

Dr Kuldeep Singh appreciated the silent work of civil society today and its rich contributions in the past in shaping up the crucial policy frame works and the legislations with respect to biodiversity and plant varieties.

Dr Kuldeep Singh appreciated todays seminar timely initiative by Dr B Sarath Babu, President of Plant Protection Association in association with public, private and civil society.



Dr Keldeep Singh mentioned that very crucial decisions are going to be taken by the government in the context and therefore the proceedings of this meeting should be handed over to the government. Thereafter a delegation must go and meet the decision making committee/government.

Dr Kuldeep Singh while agreeing with most of the issues raised by the members of civil society, he stressed the need for seed sovereignty in the context of impending amendments to the plant treaty (ITPGRFA) and the new gene bank initiative of the union government.

He specially mentioned about the disconnect among the public sector, private sector, communities, community seed banks, government of Indias initiatives and the crucial institutions in the context viz., ICAR-NBPGR, NBA and PPVFRA.

Dr Kuldeep Singh there after narrated the genesis of the so called SNGB which then was conceived as an alternative duplicate facility in the wake of risks associated with the Delhi gene bank (FNGB). He spoke passionately about lack of crucial wild species in the FNGB and the need to introduce them for posterity. Efoorts made by ICAR-NBPGR with the support of the then DG, Dr Mahapatra and the former DG Dr R S Paroda several efforts were made to identify a suitable location for establishing a national gene bank. Although several locations including J&K, central India, north east were considered but no concrete decision was taken during his stewardship at the ICAR-NBPGR.

Dr B Sarath Babu: Being the primary organizer of the seminar keeping track of the number of speakers, he made his interventions in between sessions. Adding to Dr. Abrol's observations, Dr Babu noted a critical communication gap between the public and private sectors involved in seed conservation. Although both sectors engage in similar efforts, there is very little exchange of germplasm or collaboration between them. He stressed the importance of fostering dialogue and coordination to ensure that the conservation and utilization of genetic resources serve collective interests.



During the later half the seminar, he proposed a working model second national gene bank with which all the stakeholders viz., the community seed banks involved in in-situ on-farm conservation are to be linked for a dynamic relation with the second national gene bank (SNGB). The SNGB overall management and functioning should remain with ICAR-NBPGR irrespective of the location of the proposed gene bank. Dr Babu proposed a governing body (GB) consisting of representation from the government, NBA, PPVFRA, Civil Society, Industry, SAUs, and the community Seed Banks. The governing body will oversee the functioning of the gene bank and exchange of germplasm with proper records, MTAs and the feed back information from the users. This overseeing and scrutiny of SNGB operations will facilitate the access and benefit sharing once a commercial product is produced by the user organizations.

Dr Kavitha Kuruganti: Kavitha is a social activist known for her work related to sustainable farm livelihoods and farmers' rights. She has 25 years of experience working in the development sector and is Founder Convenor of Alliance for Sustainable & Holistic Agriculture (ASHA), a pan-Indian alliance of more than 400 organisations that have come together to secure India's 'Food, Farmers & Freedom'. Kavitha has also served on several Government of India's Committees/Task Forces in an advisory role.



Dr Kuruganti spoke about plant genetic resources, their utilization, conservation, access to private institutions, community seed banks, farmers rights violations and the the acts and the international scenario concerning the exchange and benefit sharing.

- In the wake of union governments initiative, she raised basic queries on how the private sector is going to be engaged and seed sovereignty at the farmers' level.
- She raised some queries on existing gene bank about its functional status and if the entire stock is characterized or not.
- How the SNGB is going to be structures and its functional relationship with the public, private sector, communities and the community seed banks
- She has also expressed concern about the in situ conservation, sustainable farming systems and on farm diversity conservation in the context of SNGB
- She finds gaps between the general public and seed science and suggested to link the public policy with the evolving seed science in terms conservation, utilization and benefit sharing.
- She reminds the house that the question of patenting the genetic material of the plants and the living beings is yet to be adjudicated in the country. Although seeds are not patented but its genetic material is being allowed to be patented.
- She spoke lucidly about the revitalization of rainfed agriculture and seed diversity in the context of strengthening the community seed systems.

- She called for investments in the community seed systems by the state in order to save the diversity for the posterity as it has got roots in to the ancient farming systems and conservation of traditional varieties by the local farmers.
- She suggested that the entire NBPGR Database must be made accessible to the public for public scrutiny and progress. The ICAR-NBPGR (National Bureau of Plant Genetic Resources) database can be used as prior art in patent applications, particularly those related to plant genetic resources and their utilization.
- She expressed concern that there are attempts to subvert the the farmers' rights granted in the PPVFR Act ultimately taking the farming in India towards UPOV era.
- Unless the seed is handed over or made accessible to the unconditionally and incessantly to the farmers and farming communities seed sovereignty cant be achieved.

Dr Palagummi Sainath: Palagummi Sainath is widely recognized as "the voice of farmers" in India due to his extensive reporting and advocacy on issues affecting rural India, particularly farmers. He is a prominent journalist, author, and founder of People's Archive of Rural India (PARI), a platform dedicated to highlighting the stories and struggles of rural people.

Dr Sainath addressed the scientists, representatives of the civil society, industry and other public sector institutions and services. While accepting humbly about his limited knowledge in the biodiversity conservation, plant genetic resources, farmers' rights, community seed banks, national gene bank etc he spoke vividly about the distress farming community has been subjected to in India.



The decline in the number of farmers in India, alongside a growing population, presents a serious challenge to the nation's food security and livelihoods. This paradoxical situation highlights the urgent need for sustainable agricultural practices and policies to ensure the well-being of farmers and the long-term stability of the agricultural sector

Dr Sainath made the following critical and analytical observations in his talk.

- a) Millions of farmers in India are actually dissuading their children from farming as it is stressful, totally nonprofitable ending up in colossal losses driving the poor farmers often to commit suicide in the face of debt, hunger, harassment and llack of livelihood security.
- b) Dr Sainath gave factual information on millions of farmers (20millions) leaving farming since 1991until 2021 leaving food for thought to the policy makers and scientific community.
- c) He systematically narrated how farmers are declining in the country facing distress and lack of alternate livelihoods in the face of entire farming community being crushed severely leaving a note that in the days to come the gret grand children of this country may not be able to see the farmers of today living in a world without farmers.

World Café Group Discussion

Dr Divya Veluguri, Executive Director took lead in organizing the World Café Group Discussion. She divided the large group of participants in to 5 groups each consisting of 7 to 8 members and a group leader (host). The World Café method is a structured, large-group discussion technique that facilitates open and intimate conversations to explore a topic and generate collective intelligence. It involves participants rotating among small tables to discuss questions, with each table having a "host" who summarizes previous discussions and introduces new participants. The goal is to foster a dynamic and interconnected environment where ideas are shared and woven together to create a deeper understanding.

Table 1 Summary: Recommendations for a Functional National Gene Bank Model

Group 1 emphasized the need for a comprehensive review of the National Bureau of Plant Genetic Resources (NBPGR) to guide the development of a more effective second national gene bank. They recommended a structural framework that integrates both on-farm (in situ) and off-farm (ex situ) conservation methods to ensure robust biodiversity preservation. A governing body, including representatives from the government, public and private sectors, and the National Commission of Farmers, should oversee the gene bank's operations. The group proposed a unified framework for all public, private, and community seed banks, with clearly outlined roles and responsibilities, and called for a study to map the status of community seed banks and available Plant Genetic Resources (PGRs).

On financial sustainability, the group underscored the need for government and private sector contributions to fund gene bank activities. They proposed the creation of a National Gene Fund, financed through royalties paid by private entities using genetic resources. State governments should also support community seed bank activities to ensure long-term viability.

In terms of access, accountability, and ownership, the group insisted on the establishment of clear guidelines for germplasm sharing to address misuse, disease outbreaks, and other risks. They cautioned against giving private companies exclusive authority over conserved germplasm and advocated for open access to germplasm information across all types of seed banks. While the private sector may access resources, their involvement in conservation should be subject to strict public audits.

Conservation and capacity building were also highlighted, with the group recommending urgent action to protect genetic erosion hotspots and calling on NBPGR to monitor these areas routinely. Community seed banks should be equipped with training and technical support, and an incentive system should be implemented to promote on-farm conservation and uphold farmer rights.

Regarding governance, security, and accessibility, the group proposed locating the national gene bank in southern India-preferably Hyderabad or Bangalore-for better coordination and security. A public audit mechanism should be established to monitor all private and community seed banks participating in conservation. The gene bank should prioritize the domestic use of germplasm, and international sharing should be strictly limited to food security purposes, not for commercial exploitation.

Lastly, the group stressed the importance of including all types of crops in the national gene bank to ensure comprehensive biodiversity conservation. They strongly opposed the involvement of multinational corporations in conservation activities, warning that it could lead to monopolization of crucial genetic resources.

Table 3 Summary: Recommendations on Capacity Building for NBPGR and Community Seed Banks (CSBs)

Group 3 proposed a comprehensive set of recommendations to strengthen the capacity of both the National Bureau of Plant Genetic Resources (NBPGR) and Community Seed Banks (CSBs). A key recommendation was the establishment of a dedicated outreach program by NBPGR to support CSBs. This initiative should focus on building capacities in documentation skills related to local seed varieties, seed descriptors, traditional knowledge, quality assurance, and seed multiplication. Additionally, training should cover Value for Cultivation and Use (VCU) characterization and emphasize the importance of maintaining biodiversity at heritage points. The group emphasized the need for increased extension staff at NBPGR to effectively carry out these capacity-building efforts.

To foster public awareness and community engagement, the group recommended promoting community or panchayat-level diversity festivals and campaigns. Cooperative societies should be actively encouraged to establish and manage CSBs at the grassroots level. Financially, the group suggested tapping into Corporate Social Responsibility (CSR) funding and the Union Budget to support the establishment and maintenance of CSBs. Furthermore, the importance of direct support to seed farmers was highlighted as a cornerstone of sustaining local seed systems.

The group proposed the development of a centralized NBPGR Dashboard—an interoperable system to unify and streamline data across various institutions. This dashboard would provide transparency regarding seed accessions from India and abroad, clarify the linkage between the Protection of Plant Varieties and Farmers' Rights (PPVFR) Act and NBPGR, and offer access to passport data representing prior art. A National Registry should be maintained to capture standardized information from all CSBs, enabling coherent planning and policy implementation.

Enforcement of benefit-sharing mechanisms was deemed critical, and the group called on State Biodiversity Boards (SBBs), the National Biodiversity Authority (NBA), and the PPV&FR Authority to play a proactive role in this area. A governing body should be constituted to oversee these activities and ensure coordination among various stakeholders. Importantly, CSBs should be formally recognized and treated as non-commercial entities to preserve their community-oriented mission. Lastly, the group recommended that experienced farmer breeders be engaged as resource persons in Krishi Vigyan Kendras (KVKs) and agricultural universities, to transmit grassroots-level knowledge into formal research and extension systems.

Table 4 Summary: Access to Germplasm, Benefit Sharing, and Policy Concerns

Table 4 focused on the complexities surrounding access to germplasm, benefit-sharing mechanisms, and the relevant legal and policy frameworks. Ramoo initiated the discussion by emphasizing the need for a clear and inclusive model of access—proposing a shift away from exclusive licensing to a more open-source and royalty-based framework that ensures community benefit sharing through mechanisms like a gene fund. He clarified that seed banks should not merely function as warehouses; instead, they should manage borrowing through transparent systems. A key concern raised was that less than 1% of medicinal plant germplasm has ever been returned to its origin communities, underscoring deep-rooted issues in benefit distribution.

The first group highlighted the need to reconsider the private sector's role in accessing germplasm, advocating for an open-source model rather than exclusive ownership. Traceability emerged as a major issue, especially as corporates often do not disclose access details. They stressed that germplasm should be treated as a national asset under India's sovereign rights. The discussion also referenced pandemic-era practices, such as swab collection and royalty mapping for test kits, as examples of how access and benefit sharing must be structured and enforceable. Participants called for district-level bodies or commissions to ensure farmers have access to germplasm information and opposed privatisation of these national resources.

The second group focused on the shortcomings in sharing value-added profits from GI-tagged products like rice and called for benefit-sharing mechanisms in such cases. They discussed digital access to

sequencing data and proposed a subscription-based open-access system governed by the national gene bank. Participants emphasized that benefit sharing cannot follow a one-size-fits-all approach, as the stakes vary between commercial crops and rare or medicinal plants. They flagged issues like insider trading, bias among patent officers, and the need to redefine the architecture of licensing and exclusivity with greater transparency and fairness.

In the third group, participants stressed that NBPGR should expand its mandate to include activities beyond current scopes, especially given the risk of private sector dominance. They questioned whether existing benefit-sharing models are sufficient and suggested exploring alternative formulas, including perpetual licensing. The real beneficiaries, they argued, should be the seed preservers. Multiple flexible models for benefit sharing were suggested, such as monetary compensation, royalty-free access for improved derivatives, and incentive-based registrations for community seed banks. The group called for stringent legal frameworks, proper patent regulation in the context of genetic sequencing, and a unified regulatory system for access.

The fourth group reiterated the need to reinforce “license of right” provisions under the patent act, potentially with modest royalties (e.g., 4%). They emphasized that not all plant varieties should be patentable, and farmers should be granted licensing rights. Researchers should retain freedom to use tools without restrictions. Policies must reflect the nature of stakeholder engagement and differentiate between product- and service-based definitions in the non-commercial seed bank sector. Custodians of local varieties should be clearly identified for better accountability, especially since many varieties exist across multiple regions.

Finally, the fifth group proposed a hybrid model of benefit sharing, starting with a one-time payment and continuing based on a company’s earnings. They firmly stated that private companies should be kept out of the collection and conservation of genetic resources. They also raised concerns about outcrossing effects, incomplete pedigrees in patent filings, and the need for benefit sharing to reflect actual profit margins. The overarching consensus was that access to germplasm must be regulated, equitable, and transparent, and that custodians of biodiversity especially farmers and communities must be the primary beneficiaries in any sharing mechanism.

Roundtable Discussion



Dr G V Ramanjaneyulu: He emphasized that farm deliverables and kitchen needs should be synchronized and the role of Private partnership, funding, management aspects needs to be clarified by the goverment in Setup of second genebank.



Lakshamma: She emphasized that today's youth show declining interest in agriculture and stated that many prefer other forms of employment over farming. She Collects seeds from plants with strong phenotypic traits and follows traditional methods to conserve and reuse seeds for future planting seasons. She Urged that local "seed saviours" require and deserve government support to sustain their efforts. She also described the seed lending and return pattern followed in her village: Kharif Season: Seeds returned at a 1:2 ratio. Rabi Season: Seeds returned at a 1:1.5 ratio.



Nitin Ramakrishna: He mentioned that the benefit-sharing mechanism under the International Treaty on Plant Genetic Resources for Food and Agriculture (ITPGRFA) is weak. CGIAR representatives are unable to track the usage of seeds shared under the Multilateral System (MLS). Only 5 out of 25,000 users are providing feedback and participating in benefit-sharing. He emphasized the need for changes in the governance of the treaty, national genebanks, and CGIAR genebanks to facilitate the sharing of potential germplasm of edible and wild species under the MLS.

Malla Reddy: Malla Reddy stated that Telangana, and India as a whole, is self-sufficient in seed production. However, the sector is gradually being dominated—up to 80%—by corporate entities such as Monsanto, Bayer, Syngenta, Reliance, Tata, and Adani. He stressed the need for adequate research funding to develop seeds using indigenous germplasm, rather than relying on germplasm from other countries.

Dr A Prasada Rao: Forum for seed protection rights was formed between 2010-2014 in AP state. Likewise, he suggested to form a society at national level to protect seed rights. The farmers varieties needs to be focused in commercialization and business aspects like other commercial varieties. He also suggested to follow FAO framework to ensure sustainable seed development.



Donti Narasimha Reddy: He questioned the sustainability of genebanks themselves, noting that they can only function effectively with strong financial support. This is why private institutions may become involved in establishing a second genebank. The Seed Act has not been amended, largely due to the influence of private companies. While the government is now promoting the establishment of genebanks to ensure food security, these collections are often inaccessible to farmers. He cited the example of Syria, where genebank collections were relocated to Svalbard and other genebanks during the war, and the country now faces challenges in reclaiming and using those resources. This highlights the vulnerability of genebanks to socio-political disruptions. Therefore, he argued that conserving seeds in the hands of farmers, rather than relying solely on genebanks, is a more sustainable and resilient approach for seed protection and use in the country.



Haribabu: He emphasized that community seed banks and their distribution systems should be formally recognized. He also stressed the need for DNA barcoding of seed varieties to track their genetic integrity. Native varieties have been almost entirely replaced, and their protection should be a priority in the establishment of the second genebank.



Prof Dr P Raghuram: He emphasized that having a genebank is a strategic strength for the country, ensuring future access to genetic resources for all stakeholders. Similarly, community seed banks are operating at the village level; however, they receive very limited recognition and funding in India compared to other countries. He stressed the need to strengthen these local systems. He also noted that India possesses sufficient germplasm across various crops. For instance, rice includes germplasm with low nitrogen-use efficiency, yet the government continues to prioritize urea subsidies instead of investing in the development of such varieties using the rich germplasm already available. Moreover, much of the germplasm stored in genebanks remains uncharacterized. If the government fails to address these concerns, the country may face greater challenges in the event of future, unpredictable calamities.

Dr ED Israel Oliver King: He emphasized the challenges faced by community seed banks, noting that they are struggling to sustain the seed system independently. Greater focus is needed on linking these systems with market mechanisms. He advocated for the establishment of an integrated seed conservation system, with budget allocations for in situ conservation similar to those for ex situ conservation. Additionally, incentives should be provided to both individual and collective farmers involved in seed conservation. He pointed out that passport data exists for each accession in the national genebank collections, and based on this data, benefit-sharing should be ensured for the farmers.



Shri Vadde Sobhanadreeswara Rao: He suggested the gathering to come up with a simple bulletin about all these seed acts, PPVFRA, importance of national genebanks, community seed banks, access benefit sharing to create awareness among the farmers in all languages.



Krishna Ravi Srinivas, NALSAR

Krishna Ravi Srinivas highlighted the use of artificial intelligence (AI) and big data ("DAS-Data Application of Science") in this context. He referenced international declarations on the rights of rural peasants and indigenous communities, arguing that policies must go beyond agriculture to address broader social and environmental impacts on these populations.

Dr D Narasimha Reddy

The private partnership in the second genebank is really a serious concern. The entry of genebank in corporate sector is very dangerous and the government funding to community seed banks is very essential.



Prof N Venu Gopal Rao



Dr Venu Gopal addressed field-level issues related to seed conservation in cotton-growing regions. He emphasized the importance of soil health.

Dr. Kuldeep Singh

Dr. Kuldeep Singh addressed and clarified some of the views expressed by other speakers. He stated that there will be no involvement of private institutions in the establishment of the second genebank. This genebank will be a broad and inclusive structure, allowing for black-box conservation by any entity. He emphasized that the structure should be dynamic and capable of supporting the conservation of non-domesticated crop species, such as wild relatives of groundnut.

Some Observations of the civil Society

- **National Sovereignty & International Development:** The importance of balancing national sovereignty with international development goals was emphasized, especially considering interstate differences in seed systems.
- **Parliamentary Action:** It was noted that while the laws regarding peasant's rights are still not passed by the Parliament, the discussions around seed preservation have highlighted the urgent need for such legislation, with a focus on the government's role in supporting agricultural sustainability.

- **Government's Role & Corporate Influence:** There was concern about government decisions favouring private companies, urging that the focus should shift toward saving agriculture for the benefit of local farmers and communities.
- **Community Seed Savers' Perspective:** Emphasis was placed on understanding the business models, skills, and financial needs from the perspective of community seed savers and farmers. Their role is crucial in the conservation of indigenous seed varieties.
- **Germplasm Characterization & NGO Support:** The importance of characterizing germplasm was stressed. Positive examples, such as the work done by the NGO Sanjeevani, were cited as exemplary. Sanjeevani is organizing a seed festival with 2000 farmers this year to promote seed preservation.
- **Knowledge Transfer & Ecosystem Support:** The need for knowledge transfer from older to younger generations was highlighted, along with the importance of ecosystem support for seed savers and agricultural communities.
- **Gene Banks & Farmers:** It was reiterated that gene banks should not merely act as warehouses for seeds; instead, the seeds should remain with the farmers, with gene banks maintaining a list of individuals or groups who have the seeds for proper management and distribution.
- **Seed Crisis in Telangana:** Despite being a hub for seed production, Telangana is now facing a seed crisis, which underscores the urgency of addressing these issues on a national scale.
- **Future Collaboration:** The need for continued collaboration among various stakeholders, including government bodies, civil societies, and community seed banks, was emphasized to ensure a sustainable future for seed preservation.
- **Advocacy for Seed Preservation:** The advocacy efforts by civil societies for the preservation of seeds are crucial and should continue to garner support for the cause.

These concluding remarks underscore the need for urgent action in seed preservation, involving both policy changes and community-driven efforts to ensure a sustainable agricultural future.